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Original Research Article

Molecular analysis of *rpoB* gene mutation in MTB detected isolates in a tertiary care centre (AGMC) of North-East, IndiaTapan Majumdar^a, Ashmita Banik^{b,*}, Viswanadham Allada^c, Banti Das^a^a Dept of Microbiology, AGMC, Tripura, India^b Dept of Microbiology, TRIHMS, Arunachal Pradesh, India^c Govt Medical College, Srikakulam, Andhra Pradesh, India

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ABSTRACT

Background: Rifampicin (RIF), one of the first line drug in treatment of tuberculosis. It acts on *rpoB* gene which encodes RNA polymerase β subunit. In 95% of RIF resistant cases, mutations are present in *rpoB* gene. Most of them are within 81bp RIF-resistance determining region (RRDR). Xpert MTB/RIF assay has been tremendously revolutionalised the diagnosis of tuberculosis (TB). Also sequentially detect bacteria and resistance to rifampicin (rif). Approximately 96% of rif-resistant *Mycobacterium tuberculosis* (MTB) strains worldwide, showed mutations in a region at the 507-533rd amino acid residuals (81 bp) in the MTB *rpoB* gene. Here evaluation is made about frequent regions of amplification and mutation in various codons of 81bp of *rpoB* gene in rif sensitive and rif resistant cases.

Methods: A total of 4116 samples were received at Mycobacteriology laboratory, AGMC and processed in CBNAAT. Data of MTB detected samples were collected & statistically analysed to detect frequency of amplification & no amplification in various regions of 81bp of *rpoB* genes.

Results: Out of 4116 samples, MTB was detected in 1323 samples. Among them 1291 (97.58%) cases were Rif sensitive (RS) and 32 (2.41%) cases were rif resistance (RR). Most of the MTBC detected samples showed amplification in probe A then in probe C. 7.12% rif resistant cases showed mutation in either of the probe, commonest is probe E. Study also showed low bacillary loads in most of the RR cases.

Conclusion: Study highlighted variations in amplification of different regions of 81bp of *rpoB* gene in MTBC detected cases. North-east India, like other part of world, also showed highest frequency of mutation in probe E in rif resistant cases.

1. Introduction

Multidrug resistant tuberculosis (MDRTB) retards global progress of positive health, which delays achieving goal set by End TB Strategy of World Health Organisation (WHO). WHO estimated that, about half a million cases of multi-drug or rifampicin resistant TB (MDR/RR-TB) occur each year [1] (see Table 1)

RpoB gene encodes the β subunit of bacterial RNA polymerase. This gene is present in few strains of bacteria, but its length and nucleotide sequence vary between bacterial species, including mycobacteria [2]. Mutation in *rpoB* gene causes alteration in protein structure which leads to drug (rifampicin) resistance to bacteria including mycobacteria [3].

Rifampicin plays pivotal role in treatment of TB. It serves as surrogate marker for detection of MDR-TB. More than 90% of rif resistant isolates

are also isoniazid-resistant. Rifampicin binds to RNA polymerase β subunit within the DNA channel but away from active site and blocks synthesis of second and third phosphodiester bond between the nucleotides in the nucleotide backbone by steric-occlusion mechanism. This prevents elongation of the 5' end of RNA transcript thus inhibit host bacterial protein synthesis and destroy mycobacteria [4]. Majority of rif resistant mutation occur in three clusters on *rpoB* gene, cluster I, II & III encoding amino acid 507 to 533, 563–572 and 687 respectively. In *Mycobacterium tuberculosis* (MTB), majority of mutations for rifampicin resistance are in cluster-I, in 81bp hotspot core region called RRDR for “rifampicin resistance determining region” [2]. Various mutations seen in different codons of *rpoB* gene are given in table: 1 [5]. Most common mutation is in codon 531 from TCG to TGG which transform serine to leucine in amino acid chain. Tuberculosis resistance has also occurred due to mutations in the N-terminal region of *rpoB* and cluster III [2].

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Abbreviations

AGMC	Agartala Government Medical College
CBNAAT	Cartridge Based Nucleic Acid Amplification Test
INH	Isoniazid
MTB	<i>Mycobacterium tuberculosis</i>
MDR-TB	Multidrug Resistant tuberculosis
RIF	Rifampicin
R-S	Rifampicin Sensitive
R-R	Rifampicin Resistant
WHO	World Health Organisation

RpoB mutants of MTB strains also associated with mutations in rpoA or rpoC genes, encoding the RNA polymerase α or β' subunits respectively in one-third cases. This can be shown by whole genome sequencing [6]. RpoA & rpoC gene reimburses by transcriptomic alteration. Point mutation in rpoA gene leads to change in amino acids (aa) Gly31Ser, Val183Gly in aa side chain [7]. In rpoC gene changes are Glu513Gln, Val731Met, Asp735Asn, Thr845Pro, Gly986Cys. RpoC also mutated for upregulation of ribosomal protein coding genes of MTB, dysregulation of growth related essential genes, balancing the expression of arginine & glutamate synthesis associated genes in rifampicin free state. On exposure to rifampicin, rpoC mutates for upregulation of oxidative phosphorylation which was inhibited in rpoB single mutants [8].

Recently WHO endorsed Line Probe Assay (LPA) and GenoType MTBDRplus assay [9,10] for early detection of MDRTB. GenoType MTBDRplus assay detects the presence of MTB together with mutations within the rpoB gene for rifampicin (RIF) resistance and within katG gene and inhA promoter for isoniazid (INH) resistance. Cartridge Based Nucleic Acid Amplification Test (CBNAAT) along with detection of MTB also tells us about rifampicin (R) sensitivity pattern. The “hot-spots” to carry RR-DR mutations (within codons 511, 515, 516, 518, 526, 531 of rpoB genes) are used rapidly to identify RIF-resistant isolates [11,12]. CBNAAT is a multiplex real time PCR system that uses five overlapping molecular beacon probes (A-E) targeting the RRDR of rpoB gene to detect MTBC along with the mutations conferring RR. Various probes of the RRDR of rpoB gene with their codon sequence are: probe A (codons 507–511), B (codons 512–518), C (codons 518–523), D (codons 523–529) and E (codons 529–533) [13]. In most assays MTBC is detected when at least two of

Table 1
Mutations seen in rpoB gene in rif resistance cases [5].

Gene	Codons	Mutations	Change in amino acid sequences
RpoB	506	TTC→TTA	Phe→Leu
	511	CTG→CCG	Leu→Pro
	513	CAA→CCA	Gln→Pro
	515	ATG→CTG	Met→Leu
	516	GAC→TAC	Asp→Tyr
	518	AAC→GAC	Asn→Asp
	526	CAC→CAG	His→Gln
	528	CGC→TGC	Arg→Cys
	531	TCG→TGG	Ser→Trp
	572	ATC→TTC	Ile→Phe
	605	TAC→GAC	Tyr→Asp

the five probes show a cycle threshold (Ct) ≤ 38 . Any strain showing no amplification of one or more probes or when the difference between first and last Ct is more than 3.5 is considered as RR case [14].

The present study is aimed to appraise the probe mutation patterns in rifampicin resistant (R-R) cases along with amplifications in different probes of rpoB gene in both rif sensitive & rif resistant cases in a tertiary care centre of north east India based on Xpert MTB/RIF Assay.

2. Aim

To evaluate the frequency distribution of probe amplification in MTB detected rifampicin sensitive (R-S) and MTB detected rifampicin resistant (R-R) cases and probe mutation in rif resistant cases based on Xpert MTB/RIF Assay.

3. Objectives

- 3.1) To determine the frequency distribution of probe amplification in both R-S and R-R cases.
- 3.2) To assess the bacillary load in R-R cases from the cycle threshold (Ct) value of CBNAAT.

4. Methodology

It is a descriptive cross-sectional study, done in TB lab of Department of Microbiology, AGMC, Agartala, Tripura with approval of Institutional ethical committee, AGMC, with ref. No.4 (6–11)-AGMC/Medical Education/Ethics Com./2018, dated Dec 2, 2019.

4.1. Sample inclusion criteria

Pulmonary and variety of extra-pulmonary samples (eg. Pus, discharge from wound or any non-healing ulcer, gastric aspirates, broncho-alveolar lavage, bronchial brushing, ascitic fluid, pleural fluid, CSF, tissues) suspecting of presumptive TB received in last two years and two months (October 2019–December 2021) were included in the study.

4.2. Xpert MTB/RIF assay

Samples were processed as per manufacturer guidelines. This assay uses five molecular beacons (Probe A-E) with different fluorophores, encoding different regions of rpoB gene. The valid maximum cycle threshold (Ct) of 39.0 for Probes A, B, C and 36.0 for Probes D, E are set for MTB/RIF data analysis.

MTB Detected is reported when at least two probes result in Ct values within the valid range and a delta Ct min (the smallest Ct difference between any pair of probes) is < 2.0 . Rif Resistance Not Detected is reported if the delta Ct max (the Ct difference between the earliest and latest probe) is < 4.0 . Rif Resistance Detected is reported if the delta Ct max is > 4.0 .

4.3. Culture

Samples detected positive on CBNAAT were put for culture. Middle brook 7H9 liquid media was used. Samples were processed by NALC-NaOH method, inoculated in liquid media as per the protocol and incubated in MGIT 960. The culture positive samples were assessed for rif sensitivity pattern on CBNAAT and their probe amplification were analysed (Chart 1).

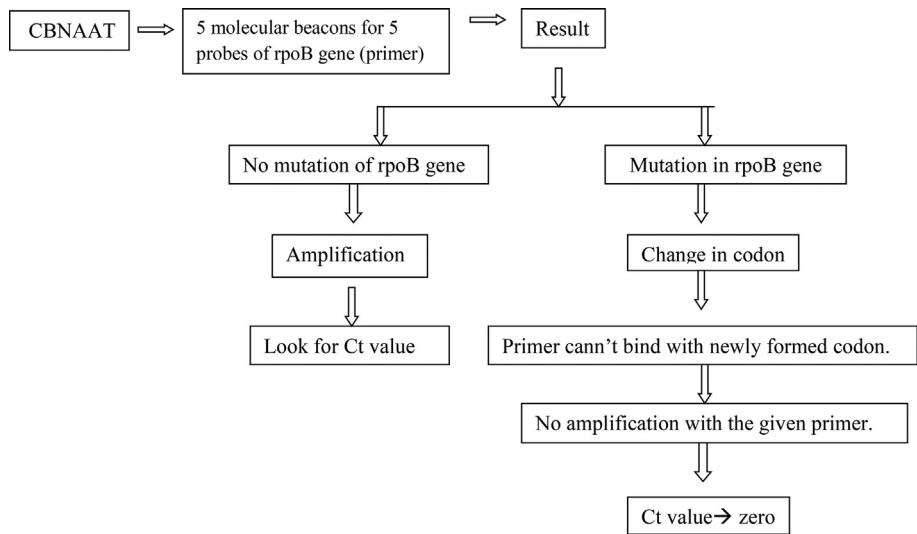


Chart 1. Algorithm for detection of mutation.

5. Results

Total of 4116 samples were processed for CBNAAT. Gene Xpert assay provided valid results in 95.45% (3929/4116) cases, error in 1.87% (77/4116), invalid results in 0.72% (30/4116), no results in 1.04% (43/4116) and indeterminate in 0.89% (37/4116) cases. Out of 3929 valid results, *Mycobacterium tuberculosis* (MTB) was detected in 33.67% (1323/3929) and not detected in 66.33% (2606/3929) samples. MTB detected samples were put on culture followed by MPT64 testing for further confirmation. Among the culture positive samples, 97.58% (1291/1323) were rifampicin sensitive on CBNAAT and 2.41% (32/1323) were rifampicin resistant on CBNAAT.

The frequency distribution of amplification in various region of rpoB gene among MTB detected Rif sensitive cases were compiled and statistically analysed. Among 400 randomly selected samples, in 332 (83%)

samples probe A amplified earlier followed by probe C (15.5% = 62/400), B (2.5% = 10/400), D (0.75% = 3/400) & E (0.25% = 1/400). Second probe to be amplified in descending order of their frequency were probe C (323/400 = 80.75%), A (70/400 = 17.5%), D (4/400 = 1%), B (3/400 = 0.75%) (Fig. 1).

Among 32 Rifampicin resistant cases, various probes in decreasing order of frequency for early amplification were probe A > probe C > probe B and for second early amplification were probe C > probe B = probe A > probe D. However there is no amplification or least number of amplification in probe D and probe E.

25 out of 32 (25/32 = 78.12%) rif resistant samples showed no amplification (i.e mutation) in either of the probe. Among them 72% (18/25) cases showed mutation in probe E, 12% (3/25) in probe D, 8% (2/25) in B, 4% (1/25) in A & C. Remaining 7 samples (7/32 = 21.87%) showed a difference of >10 between maximum and minimum Ct value (Table 2, Fig. 2).

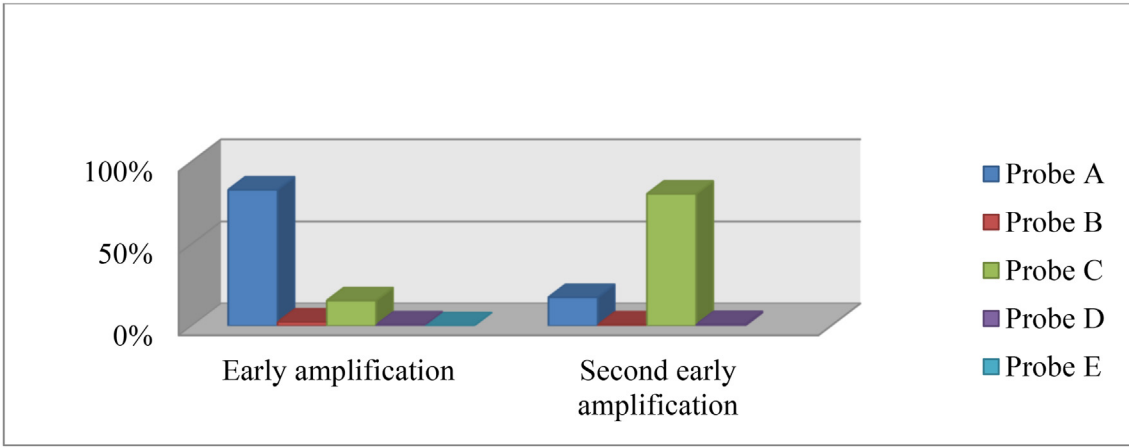


Fig. 1. Frequency distribution of early amplification of various probes in rif sensitive cases.

Table 2
Frequency distribution of mutation in various probes in rif resistant cases (n = 32).

	Mutation					No Mutation All probe
	Probe A	Probe B	Probe C	Probe D	Probe E	
RR cases	25	25	25	25	25	7
No mutation detected	1	2	1	3	18	7
Proportion	4%	8%	4%	12%	72%	21.87

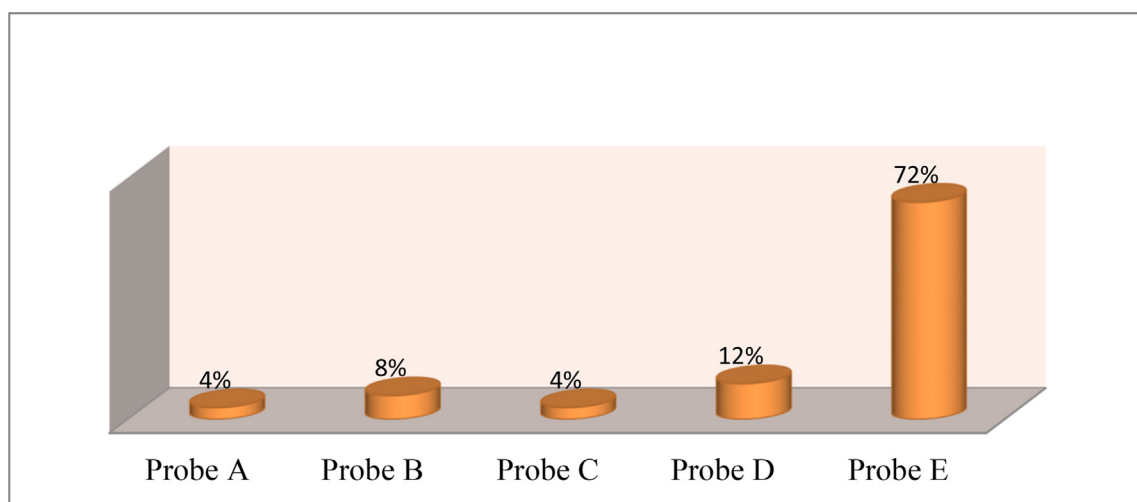


Fig. 2. Frequency distribution of mutation in various rif resistant cases.

Roy I in his study defined the concentration of bacilli in CBNAAT by Ct range (>28 = very low, $22-28$ = low, $16-22$ = medium and <16 = high) [20]. This study also determined the bacillary load of MTB based on cycle threshold (Ct) value of CBNAAT. Ct values of all the MTB detected samples which were culture positive also were compiled and classified in four groups (Table 3). Samples showing lower Ct value had higher bacillary load and the samples showing higher Ct value had the lower bacillary loads.

Among 32 R-R samples 40.63% (13/32) were having very low bacillary load, 25% (8/32) low bacillary loads, 21.87% (7/32) medium bacillary loads and 12.5% (4/32) high bacillary load. However no difference in mutation of *rpoB* gene was seen in pulmonary and extra-pulmonary samples. The study was limited to the frequency distribution of *rpoB* gene of rifampicin in MTB detected cases. To see the spectrum of demographic variations, further study is needed.

6. Discussion

Proportion of presumptive TB among suspected cases attending TB lab, Dept of Microbiology, AGMC is 33.67%. With large number of cases, it is easier to elaborate other factors associated with diagnosis of TB. This study tells us about the site of mutation on *rpoB* gene in R-R cases in a tertiary care centre, in Tripura. In Mizoram, a study was done from Dec 2014 to May 2021 on 2894 TB cases detected via. Xpert MTB/RIF assay. The most common RRDR *rpoB* gene mutations in the 81 bp were in codons 507–511 (25.9%), followed by codons 529–533 (23.5%), 523–529 (9.8%), 511–518 (2.6%), 518–523 (0.2%) which corresponds to probes A, E, D, B and C, respectively [15].

Study showed early amplification in probe-A and probe-C in both MTB detected RS & RR cases and late or no amplification (mutation) in probe E and D in MTB detected RR cases. Previous studies in India described common *rpoB* mutations in MTB detected cases on codon 516 region of probe-B, codon 526 region of probe-D [16].

Table 3

Classification of bacillary load based on Ct value.

Cycle threshold value*	Bacillary load
Less than or equal to (14.33 ± 1)	High bacillary load
(14.33 ± 1) to (19.93 ± 1)	Medium bacillary load
(19.93 ± 1) to (24.74 ± 1)	Low bacillary load
(24.74 ± 1) to (30.28 ± 1)	Very low bacillary load

Study also showed 72% of mutation in probe E in RR cases in CBNAAT. Reddy et al. in their study showed 531 codon located in Probe E region as predominant mutations in R-R cases [13]. This coincide with our study. Mutation in codon 531 (TCC→TGG) causes substitution of serine to tryptophan. This describes molecular feature of *rpoB* gene in R-R cases. Rahman A et al. in their study described gene Xpert as a quantitative method, which predicts resistance based on Ct value and it is possible to get variable result than any qualitative method like LJ-DST. However most frequent mutation is in codon 531, which varies from 20 to 71% [17–19].

As earliest amplification was in probe A, then in probe C in maximum samples in both R-S and R-R cases. So codon sequence of *rpoB* gene for early amplification in both R-S and R-R cases are similar. However 78.12% (25/32) of total R-R cases showed mutation in either of the probe, highest frequency of mutation was in probe E (72%), second highest in probe D (12%) and 7 out of 32 samples (21.87%) showed a difference in Ct value of >10 between the highest and lowest Ct of that sample which is just similar to the molecular findings in other region of the country [16].

The bacillary load of MTBC in R-R cases were analysed. It was observed that majority of rif resistant samples were of very low bacillary load with Ct value ranging from 24.74 ± 1 to 30.28 ± 1 and this is a rare finding.

7. Conclusion

This study showed the frequency distribution of amplification in MTB detected rifampicin sensitive and resistant cases and mutation in rifampicin resistant cases to get vivid picture of the molecular pattern of *rpoB* gene in various situations. Majority of rifampicin resistant samples showed mutation among codon 529 to 533. Xpert assay may be a convenient tool to investigate molecular pattern of *rpoB* gene in resource inadequate set up. However researchers must look for more advanced diagnostic methods to identify the molecular pattern of drug resistant TB elaborately.

Ethical statement

Study was done in TB lab, Dept of microbiology, AGMC, Agartala, Tripura (west) along with thesis work with approval of Institutional ethical committee, AGMC, with ref. No.4 (6–11)-AGMC/Medical Education/Ethics Com./2018, dated Dec 2, 2019.

Conflicts of interest

None.

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